Regression for Spatial Models with Interference

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Abstract

We consider the problem of inference in linear regression models with interference. In our setting with multiple locations and multiple sectors, common trade shocks generate differential exposure among locations, spillovers between locations, and heterogeneous responses in the outcome. When the data generating process (DGP) admits an infeasible representation of these outcomes in terms of a set of sufficient statistics for the spillovers, we can instead incorporate interference into regressions using own exposure and a function of neighbor exposure and an interactions matrix, although this may lead to misspecification of the regression model for the true underlying DGP. We show that the estimands of the linear model can be expressed as linear combinations of sector-location treatment effects, or location treatment effects. Despite this misspecification, we provide conditions for (i) the OLS estimators to be consistent for the estimands, and (ii) valid inference using a variance estimator that agrees with a class of structural models of gravity trade and produces confidence intervals with asymptotically correct coverage. In contrast to cluster inference methods used in the empirical literature, which assume independence between groups of locations, we show that our conditions allow for a single cluster of locations exhibiting network dependence, but require the total misspecification error to grow at a rate smaller than the sample size.

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